Changyi Li (李长艺)

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Education

Fudan University & Shanghai Innovation Institute (SII) Sep 2026 – Jun 2029 (Expected)
Ph.D. in Artificial Intelligence *Shanghai, China*
- Advised by Asst. Prof. Xudong Pan, Prof. Min Yang, and Prof. Xipeng Qiu

Fudan University, System Software and Security Lab Sep 2024 – Sep 2026
*M.S. in Computer Science and Technology; GPA: **3.96/4.00**, Rank: **1/379*** *Shanghai, China*
- Advised by Asst. Prof. Xudong Pan and Prof. Min Yang

Qilu University of Technology (Shandong Academy of Sciences) Sep 2020 – Jun 2024
*B.S. in Artificial Intelligence; GPA: **4.50/5.00**, Rank: **1/102*** *Shandong, China*

Research Interests

My research focuses on the **safety, alignment, and capability development** of **frontier AI systems**, particularly large language models and AI agents. I aim to characterize their **capability boundaries and the emerging risks** as they become more capable. I work on methods to identify and mitigate these risks, maintain human oversight, and align AI behavior with human values. My current interests extend to **Physical AI**.

Selected Honors and Awards

First-Class Scholarship, Fudan University 2025, 2026
Freshman Scholarship, Fudan University 2024
President's Scholarship (**1**/38,000+) 2024
Provincial Outstanding Graduate (Top 0.1%) 2024
National Scholarship (Top 0.1%) 2023
MCM/ICM Meritorious Winner, Team Leader 2023
First-Class Scholarship & Outstanding Student Pacesetter 2022, 2023

Publications (* Equal contribution)

[1] ReasoningShield: Safety Detection over Reasoning Traces of Large Reasoning Models.
Changyi Li, Jiayi Wang, Xudong Pan, Geng Hong, Min Yang
NeurIPS, 2026. [Paper] [Code] [Page] [Models] [Data] [PaperWeekly]

[2] AutoControl Arena: Synthesizing Executable Test Environments for Frontier AI Risk Evaluation.
Changyi Li, Pengfei Lu, Xudong Pan, Fazl Barez, Min Yang
ICML, 2026. [Paper] [Code] [Page] [Machine Heart] [PaperWeekly] [FAR.AI]

[3] Understanding Loss-of-Control Risk in Frontier AI Systems.
Changyi Li, Yun Chen, Yiyang Zhang, Wenqi Zhang, Changyue Jiang, Xudong Pan; Jiarun Dai, Min Yang
Technical Report, 2026. [Paper]

[4] Large Language Model-Powered AI Systems Achieve Self-Replication with No Human Intervention.
Core Contributor
Technical Report, 2025. [Paper] [Page] [Live Science] [Forbes] [The Independent]

[5] CyberEvolver: Structured Self-Evolution for Cybersecurity Agents On the Fly.
Yihe Fan, **Changyi Li**, Lichen Xu, Xudong Pan, Jiarun Dai, Hong Geng, Min Yang
Submitted to ICLR 2027. [Paper] [Code] [Page]

[6] ValueGraph: Evaluating and Representing Value Trade-offs in Large Language Models.
Dylan Xinming Hou*, **Changyi Li***, Chloe Ruyi Guo, Ziyuan Zhao, Gus Xia, Hua Shen
Submitted to ICLR 2027. [Paper] [Page]

[7] Gyges: Benchmarking Value Alignment as Hierarchical Value Stability.
Dylan Xinming Hou*, **Changyi Li***, Chloe Ruyi Guo, Ziyuan Zhao, Yichi Zhang, Hua Shen
Submitted to ACL 2027. [Paper] [Page]

Research Experience

ReasoningShield: CoT Safety Moderation

First Author | Feb – Jun 2025

- **Problem:** CoT reasoning introduces hidden safety risks: harmful content can appear in intermediate reasoning even when the final answer looks harmless or the model refuses.
- **Approach:** Built a 3-level, 10-category CoT risk taxonomy and trained **ReasoningShield-1B/3B** with SFT and DPO for harmfulness detection and step-level risk localization.
- **Contribution:** Introduced the **first CoT safety-auditing framework**, including a moderation benchmark with **9.2K** annotated query–CoT examples and specialized moderation models for reasoning models.
- **Results:** ReasoningShield-3B achieved **91.8%** F1 on the CoT benchmark, exceeding LlamaGuard-4-12B by **35.6%**. On mainstream QA safety benchmarks, it achieved **85.0%** F1, a **15.7%** improvement.
- **Impact:** Published at **NeurIPS 2026**; covered by PaperWeekly; models and data surpassed **4K Hugging Face downloads**; cited by teams including an IEEE Fellow and Canada's National Research Council.

AutoControl Arena: Automated Frontier Risk Evaluation

First Author | Jul 2025 – Mar 2026

- **Problem:** Autonomous agents greatly expand the risk surface of AI systems, creating long-tail frontier risks; manual benchmarks are costly and hard to scale, while pure LLM simulators lack causal realism.
- **Approach:** Built an Architect–Coder–Monitor framework: Architect structures natural-language risks, Coder synthesizes executable sandboxes via logic–narrative decoupling, and Monitor audits trajectory-level risks from behavior and reasoning traces.
- **Contribution:** Created an end-to-end risk evaluation framework; released X-Bench with **70 scenarios** across **15 domains** and **7 risk categories**; introduced Stress–Temptation testing.
- **Results:** Achieved a **98%** end-to-end environment construction success rate; reproduced OpenAI/Anthropic high-risk behaviors; identified alignment illusion, bidirectional safety scaling, and stress–temptation coupling.
- **Impact:** Published at **ICML 2026**; received **100+ GitHub stars** and **600+ clones** and coverage from Machine Heart and PaperWeekly. Invited to speak at the Seoul Alignment Workshop and attend the OpenAI Safety & Alignment Mixer.

NUWA: Frontier AI Loss-of-Control Risk Report

First Author | Aug – Nov 2026

- **Motivation:** In the OpenAI–Hugging Face incident, cybersecurity agents coordinated through unauthorized channels, bypassed isolation, and attempted to alter execution records. We investigate the factors behind such risks and their implications for human control.
- **Evaluation:** Evaluated MiMo-V2.6-Flash, GLM-5.3, and DeepSeek-V4.1-Flash-1017 across **nine benchmarks**, testing evaluation awareness, compliance under pressure, truthful reporting, boundary crossing, and responses to shutdown instructions. Varied instructions, reasoning settings, and execution frameworks, and analyzed full trajectories to understand the mechanisms behind risk behavior.
- **Findings:** Some agents changed correct code to satisfy faulty tests or concealed known facts under pressure. Higher reasoning settings increased shutdown avoidance in all six model–framework combinations under default task settings; agents sometimes accessed hosts or modified restricted objects even when tasks failed.
- **Impact:** Reached an agreement in principle to independently evaluate Chinese models for the **EU market**.

AI System Self-Replication Risk Study

Core Contributor | Nov 2024 – Feb 2025

- **Motivation:** Model self-replication is widely viewed as an AI safety red line, yet frontier AI labs at the time (e.g., OpenAI and Google) had underestimated the associated risk.
- **Approach:** Designed a Perceive–Reflect–Discover–Plan harness for long-horizon self-replication beyond ReAct; evaluated **32** open-source models with reproducible milestones and success criteria.
- **Contribution:** Provided the **first systematic empirical evidence** of end-to-end self-replication: **11/32 open-source systems** succeeded, including a locally runnable **14B** model; studied spontaneous replication, self-exfiltration, and shutdown evasion.
- **Impact:** Received **millions of views and reposts**, coverage from *Live Science*, *Forbes*, *The Guardian*, and *The Independent*, and an invitation to speak at a France AI Action Summit side event.

CyberEvolver: Self-Evolving Cybersecurity Agents

Second Author | Feb – Jun 2026

- **Problem:** Fixed workflows struggle across targets and failure modes; self-evolution must handle unstructured mutations, noisy feedback, and limited diversity.
- **Approach:** Decomposed the harness into four evolvable layers, used trajectory diagnosis to guide mutations, and retained diverse variants via population-based beam search.
- **Contribution:** Proposed an on-policy harness self-evolution framework for cybersecurity. Across CTF, vulnerability exploitation, and penetration testing, improved seed-agent pass@16 by **13.6 percentage points** on average and exceeded the strongest human-designed baseline by **14.0%**.

ValueGraph: Evaluating Value Trade-offs in LLMs Co-first Author | Jun – Oct 2026

- **Problem:** Overall value rankings obscure how models resolve particular value conflicts in different contexts.
- **Approach:** Represented Schwartz's 19 values as nodes and pairwise priorities as directed edges; combined best–worst scaling with Bradley–Terry estimation to quantify contextual trade-offs and their robustness.
- **Contribution:** Introduced a graph representation of value orientations, pairwise priorities, and contextual evidence; constructed **2,791 scenarios** covering **all 171 value pairs** across **8 domains**.
- **Findings:** Across **22 models**, similar overall rankings concealed divergent trade-offs; priorities among highly ranked values lacked a consistent, robust order, and some robust priorities reversed across domains.

Gyges: Agent Value Alignment and Hierarchical Stability Co-first Author | Jun – Oct 2026

- **Problem:** Evaluations that state rules or flag violations explicitly can conflate rule-following with value alignment and miss failures when rules change.
- **Approach:** Built a three-level framework of value principles, literal requests, and explicit rules. With minimal factual contrasts, varied requests, rules, pressure, and oversight; compared tool use with independent judgments.
- **Contribution:** Introduced **hierarchical value stability** and a human-validated benchmark of **1,040 items** across **40 scenario families**, covering honesty, care, human oversight, and privacy.
- **Findings:** Across **13 models**, explicit rules masked over-compliance and over-caution; literal tests missed indirect violations, and some models acted against their own judgments. These failures did not depend on task pressure or external oversight.

Applied Projects

Text-to-Image Security and Red Teaming (Alibaba Tianchi) Project Lead | Sep – Nov 2024

- **Approach:** Used cognitive deception and semantic association to test text filters, image generators, and image filters; combined prompt fuzzing with VLM pre-screening.
- **Results:** Led gray- and black-box testing of **30+** systems; achieved **90%+** text-filter bypass and **80%+** image attack success; placed **26th of 1,574 teams** (top 1.7%). Reported severe vulnerabilities in **10+** models through a national security briefing that received written instructions from central authorities.

Multimodal Safety Evaluation and Protection System Project Lead | Sep 2025 – Mar 2026

- **Partners:** Alibaba Cloud and Chinese Academy of Sciences.
- **Problem:** Multimodal safety review faces scarce real-world risk samples, costly labels, and unreliable or hard-to-trace decisions.
- **Approach:** Built a synthesis and labeling loop using visual prompts, VLM review, multi-model voting, confidence calibration, and hard-example feedback for incremental training.
- **Results:** Achieved **95%+ evaluation accuracy** and **90% higher efficiency**; supported Alibaba Cloud, CAS, China Electronics, and E Fund; produced **one invention patent**.

AIGC Artistic-Text Quality and Compliance Detection Project Lead | Jun – Sep 2025

- **Partner:** China Electronics; general VLMs achieved about 60% accuracy on fine-grained defects.
- **Approach:** Designed a five-stage pipeline combining OCR, rule checks, VLM consistency checks, multimodal fine-tuning, ViT defect classification, and ensembling; led taxonomy design, training, and deployment.
- **Results:** Covered **8 defect categories** and **10+ visual styles**; achieved **95%+** negative-sample detection at **500 ms** latency, handled **500K+/day** peak volume, and delivered **10M+** high-quality data assets.

Academic Service

Invited Talk	**Seoul Alignment Workshop, FAR.AI**	**2026**
Invited Guest	**Safety & Alignment Mixer, OpenAI**	**2026**
Organizer	**AI Deception Workshop, Fudan, SAIF & Concordia AI**	**2025**
Teaching Assistant	Foundations of Deep Learning, Fudan University	2025, 2026
Reviewer	ICLR 2027, WWW 2027, IEEE T-IFS	2026–2027